

Title Slide

Predicting SpaceX Launch Success

A Data Science Capstone Project

Your Name | Date

Executive Summary (1 pt)

Goal: Predict SpaceX launch success.

Steps: Collection, cleaning, EDA, visualization, modeling.

Key Result: 92%+ accuracy.

Introduction (1 pt)

SpaceX's mission: affordable, reliable space travel.

Prediction helps mission planning & risk reduction.

Data Collection Methodology (1 pt)

4.1 SpaceX API (with flowchart + GitHub)

4.2 Web Scraping (with flowchart + GitHub)

4.3 Data Wrangling (cleaning + features + GitHub)

EDA & Visual Analytics Methodology (3 pts)

5.1 Charts: scatter, bar, line (+ GitHub)

5.2 SQL EDA: queries, results (+ GitHub)

5.3 Interactive: Folium + Dash (+ GitHub)

Predictive Analysis Methodology (1 pt)

Models: KNN, Logistic Regression, Decision Tree, SVM

Metrics: Accuracy, Recall, AUC

Hyperparameter tuning (+ GitHub)

EDA Visualization Results (6 pts)

1. Flight No vs Site
2. Payload vs Site
3. Success vs Orbit
4. Flight No vs Orbit
5. Payload vs Orbit
6. Success trend line

EDA with SQL Results (10 pts)

1. Unique Sites
2. Sites 'CCA'
3. Payload Mass
4. Avg Payload F9
5. 1st Ground Landing
6. Drone Ship 4000-6000
7. Outcome Counts
8. Max Payload Booster
9. Launches 2015
10. Ranked Success

Interactive Map with Folium (3 pts)

1. Map markers
2. Records per site
3. Proximity distances (rail, road, coast)

Plotly Dash Dashboard (3 pts)

1. Pie: All Sites Success
2. Pie: Top Site Success
3. Scatter: Payload vs Outcome

Predictive Analysis Results (6 pts)

1. KNN
2. Logistic Regression
3. Decision Tree
4. SVM
5. Best Model
6. Confusion Matrix

Conclusion (1 pt)

EDA: trends & visuals

SQL: payload & site insights

Interactive tools

Best Model: Random Forest 92%

Creativity & Innovation (1 pt)

Animations, infographics, extra charts, clean design.

Innovative Insights (1 pt)

Cost optimization

Reliable boosters

Payload/orbit efficiency

Enhanced dashboard.

References & GitHub

GitHub: [Your Link Here]

Data: SpaceX API, Kaggle